

Image Analysis Report

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Source	corrosion.jpg
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Executive Summary

This image displays severe localized corrosion and complete coating breakdown along a horizontal weld seam, consistent with a high severity rating. The presence of heavy exfoliating scale indicates significant metal loss and potentially compromised structural integrity of the joint.

Key Findings

#	Finding
1	Significant localized corrosion centered on a horizontal weld or joint interface.
2	Formation of heavy, brittle iron oxide scale (exfoliation) in the central area.
3	Complete failure of the protective coating system in affected zones.
4	Visible under-film corrosion creeping beneath the remaining paint edges.
5	Distinct vertical rust bleeding patterns suggesting active moisture ingress and corrosion.
6	Presence of pitting clusters in the immediate vicinity of the main corrosion site.

Localized detections (VLM, approximate)

- Severe Exfoliation (conf. 92%): x=0.340, y=0.120, w=0.400, h=0.720
- Corroded Weld Seam (conf. 88%): x=0.000, y=0.440, w=1.000, h=0.140

Recommendations

#	Recommendation
1	Schedule immediate abrasive blast cleaning (SSPC-SP 10) to expose the bare metal substrate for assessment.
2	Perform Ultrasonic Thickness (UT) measurements to quantify the depth of pitting and overall wall loss.
3	Evaluate the structural integrity of the weld using Magnetic Particle Inspection (MPI) after cleaning to check for cracks.
4	Determine if remaining wall thickness is below minimum allowable (T-min) per engineering standards.
5	Implement a repair solution such as a weld overlay or flush patch if structural limits are exceeded.
6	Re-coat the surface using a high-performance system including a zinc-rich primer and epoxy mid-coat.